

Here's a compact, self-contained Hilbert-style proof system for your recognition/comprehension logic, plus a matching sequent calculus.

1. Language

- Atoms: $\backslash(p,q,r,\text{dots}\backslash)$, plus fixed atoms per entity $\backslash(e\backslash)$:
 - $\backslash(\text{Model}\backslash e, \text{Predict}\backslash e, \text{Track}\backslash e\backslash)$
 - $\backslash(\text{SurpriseHigh}\backslash e, \text{SurpriseLow}\backslash e, \text{Notice}\backslash e\backslash)$
- Modal operators (per entity $\backslash(e\backslash)$):
 - $\backslash(\Box^{\text{Rec}}\backslash e, \Diamond^{\text{Rec}}\backslash e\backslash)$
 - $\backslash(\Box^{\text{Comp}}\backslash e, \Diamond^{\text{Comp}}\backslash e\backslash)$
- Temporal operators (minimal):
 - $\backslash(P\varphi\backslash)$: " $\backslash(\varphi\backslash)$ held at the previous moment"
 - $\backslash(F\varphi\backslash)$: " $\backslash(\varphi\backslash)$ will hold at some future moment"

Formulas are built from atoms with $\backslash(\text{neg}, \text{land}, \text{lor}, \text{rightarrow}\backslash)$, the two $\backslash(\Box\backslash)$ operators, and $\backslash(P,F\backslash)$.

2. Hilbert-style system

2.1 Propositional core

Axioms (all schemas):

- P1: $\backslash(\varphi \rightarrow \psi \rightarrow \varphi)\backslash$
- P2: $\backslash((\varphi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \chi)))\backslash$
- P3: $\backslash((\text{neg } \varphi \rightarrow \text{neg } \psi) \rightarrow (\psi \rightarrow \varphi))\backslash$

Rule:

- MP (Modus Ponens): from $\backslash(\varphi\backslash)$ and $\backslash(\varphi \rightarrow \psi)\backslash$, infer $\backslash(\psi)\backslash$.

2.2 Modal part (normal K for each box)

For each entity $\backslash(e\backslash)$:

Axioms:

- K-Rec($_e$):

$$\begin{array}{l} \backslash[\\ \Box^{\{\text{Rec}\}}e(\varphi \rightarrow \psi) \rightarrow (\Box^{\{\text{Rec}\}}e \varphi \\ \rightarrow \Box^{\{\text{Rec}\}}_e \psi) \\ \backslash] \end{array}$$

- K-Comp($_e$):

$$\begin{array}{l} \backslash[\\ \Box^{\{\text{Comp}\}}e(\varphi \rightarrow \psi) \rightarrow (\Box^{\{\text{Comp}\}}e \varphi \\ \rightarrow \Box^{\{\text{Comp}\}}_e \psi) \\ \backslash] \end{array}$$

Rules:

- Nec-Rec(e): from $\vdash \varphi$, infer $\vdash \Box^{\{\text{Rec}\}}e \varphi$.
- Nec-Comp(e): from $\vdash \varphi$, infer $\vdash \Box^{\{\text{Comp}\}}e \varphi$.

(You can add T, 4, 5, etc., if you want frame properties; not required for the conceptual split.)

2.3 Temporal part (minimal)

Axioms:

- Monotonicity of $\langle F \rangle$:

$$\begin{array}{l} \backslash[\\ (\varphi \rightarrow \psi) \rightarrow (F\varphi \rightarrow F\psi) \\ \backslash] \end{array}$$

- Monotonicity of $\langle P \rangle$:

$$\begin{array}{l} \backslash[\\ (\varphi \rightarrow \psi) \rightarrow (P\varphi \rightarrow P\psi) \\ \backslash] \end{array}$$

Rules:

- From $\vdash \varphi \rightarrow \psi$, infer $\vdash F\varphi \rightarrow F\psi$.
- From $\vdash \varphi \rightarrow \psi$, infer $\vdash P\varphi \rightarrow P\psi$.

2.4 Non-logical axioms (recognition vs comprehension)

For each entity e :

Comprehension constraints:

- $C1(e): \Box^{\text{Comp}} e \top \rightarrow \text{Model}_e$
- $C2(e): \Box^{\text{Comp}} e \top \rightarrow \text{Predict}_e$
- $C3(e): \Box^{\text{Comp}} e \top \rightarrow \text{Track}_e$
- $C4(e): \Box^{\text{Comp}} e \top \rightarrow \text{SurpriseLow}_e$

Recognition constraints:

- $R1(e): \Box^{\text{Rec}} e \top \rightarrow (\neg \text{Model}_e \vee \neg \text{Predict}_e)$
- $R2(e): \Box^{\text{Rec}} e \top \rightarrow (\text{SurpriseHigh}_e \vee \neg \text{Track}_e)$

Mutual incompatibility (optional):

- $M1(e): \Box^{\text{Comp}} e \top \rightarrow \neg \Box^{\text{Rec}} e \top$

Noticing invariant:

- $N1(e): \text{Notice}_e \rightarrow \text{SurpriseHigh}_e$
- $N2(e): \text{Notice}_e \rightarrow P(\neg \Box^{\text{Comp}} e \top)$

Optional link (no noticing under low surprise):

- $S1(e): \text{SurpriseLow}_e \rightarrow \neg \text{Notice}_e$

From $C4(e)$ and $S1(e)$, you can derive in this system:

$$\Box^{\text{Comp}} e \top \rightarrow \neg \text{Notice}_e$$

i.e. in comprehension mode, you don't "notice" e —you're already tracking it.

3. Sequent calculus system

We mirror the same logic with sequents $\Gamma \rightarrow \Delta$, where Γ, Δ are finite multisets of formulas.

3.1 Structural rules

- Identity: $\varphi \rightarrow \varphi$
- Weakening (left/right):
 - from $\Gamma \rightarrow \Delta$, infer $\Gamma, \varphi \rightarrow \Delta$
 - from $\Gamma \rightarrow \Delta$, infer $\Gamma \rightarrow \Delta, \varphi$
- Contraction (left/right):
 - from $\Gamma, \varphi, \varphi \rightarrow \Delta$, infer $\Gamma, \varphi \rightarrow \Delta$
 - from $\Gamma \rightarrow \Delta, \varphi, \varphi$, infer $\Gamma \rightarrow \Delta, \varphi$
- Cut:
 - from $\Gamma \rightarrow \Delta, \varphi$ and $\Sigma, \varphi \rightarrow \Lambda$, infer $\Gamma, \Sigma \rightarrow \Delta, \Lambda$

3.2 Propositional rules (standard)

Examples (others analogous):

- $\wedge$ -Left:

$$\frac{\Gamma, \varphi, \psi \rightarrow \Delta}{\Gamma, \varphi \wedge \psi \rightarrow \Delta}$$
- $\wedge$ -Right:

$$\frac{\Gamma \rightarrow \Delta, \varphi \quad \Gamma \rightarrow \Delta, \psi}{\Gamma \rightarrow \Delta, \varphi \wedge \psi}$$
- $\rightarrow$ -Left:

$$\frac{\Gamma \rightarrow \Delta, \varphi \quad \Sigma, \psi \rightarrow \Lambda}{\Gamma, \Sigma, \varphi \rightarrow \psi \rightarrow \Delta, \Lambda}$$
- $\rightarrow$ -Right:

$$\frac{\Gamma, \varphi \rightarrow \Delta, \psi}{\Gamma \rightarrow \Delta, \varphi \rightarrow \psi}$$

Negation, $\neg$ rules as usual.

3.3 Modal rules (for each $\Box^{\text{Rec}}$, $\Box^{\text{Comp}}$)

Use a standard K-style rule schema. For each box $\Box^*$ (either $\Box^{\text{Rec}}$ or $\Box^{\text{Comp}}$):

- Box-Right (K):

$$\frac{\Gamma \rightarrow \Delta, \varphi}{\Box^* \Gamma \rightarrow \Box^* \Delta, \Box^* \varphi}$$

where $\Box^* \Gamma$ means prefix every formula in Γ with $\Box^*$, similarly for $\Box^* \Delta$.

- Box-Left (optional, often derivable in labelled systems):

$$\frac{\Gamma, \varphi \rightarrow \Delta}{\Gamma, \Box^* \varphi \rightarrow \Delta}$$

(This is sound in many labelled or relational presentations; you can also encode modal behavior via labels and relational rules instead.)

If you prefer a labelled sequent calculus, you'd have sequents of the form:

$$w : \Gamma \rightarrow w : \Delta$$

and rules that explicitly manipulate accessibility $(w R^* v)$. But for brevity, I'm keeping it schematic.

3.4 Temporal rules (for P, F)

Minimal monotonic rules:

- F-Right:

$$\frac{\Gamma \rightarrow \Delta, \varphi}{\Gamma \rightarrow \Delta, F\varphi}$$

- F-Monotonicity (admissible): if $\Gamma \rightarrow \varphi \rightarrow \psi$ is derivable, then so is $\Gamma \rightarrow F\varphi \rightarrow F\psi$.

Similarly for P :

- P-Left:

$$\frac{}{P\varphi \rightarrow \varphi}$$

$\frac{\Gamma, \varphi \rightarrow \Delta}{\Gamma, P \rightarrow \Delta}$
 $\downarrow$

- P-Monotonicity: from $(\rightarrow \varphi \rightarrow \psi)$, derive $(\rightarrow P \rightarrow \rightarrow P \psi)$.

(Again, a full temporal sequent calculus could be more elaborate; this is the minimal skeleton to support your invariant.)

3.5 Domain-specific axioms as initial sequents

For each entity (e) , add these as axiom sequents (always available):

- C1 (e) : $(\rightarrow \Box^{\text{Comp}} e \rightarrow \text{Model}_e)$
- C2 (e) : $(\rightarrow \Box^{\text{Comp}} e \rightarrow \text{Predict}_e)$
- C3 (e) : $(\rightarrow \Box^{\text{Comp}} e \rightarrow \text{Track}_e)$
- C4 (e) : $(\rightarrow \Box^{\text{Comp}} e \rightarrow \text{SurpriseLow}_e)$
- R1 (e) : $(\rightarrow \Box^{\text{Rec}} e \rightarrow (\neg \text{Model}_e \vee \neg \text{Predict}_e))$
- R2 (e) : $(\rightarrow \Box^{\text{Rec}} e \rightarrow (\text{SurpriseHigh}_e \vee \neg \text{Track}_e))$
- M1 (e) : $(\rightarrow \Box^{\text{Comp}} e \rightarrow \neg \Box^{\text{Rec}} e \rightarrow \top)$
- N1 (e) : $(\rightarrow \text{Notice}_e \rightarrow \text{SurpriseHigh}_e)$
- N2 (e) : $(\rightarrow \text{Notice}_e \rightarrow P(\neg \Box^{\text{Comp}} e \rightarrow \top))$
- S1 (e) : $(\rightarrow \text{SurpriseLow}_e \rightarrow \neg \text{Notice}_e)$

From these, you can derive in the sequent system the same key theorem:

$\downarrow$
 $\rightarrow \Box^{\text{Comp}} e \rightarrow \neg \text{Notice}_e$
 $\downarrow$

and, using N2 (e) , that noticing implies prior non-comprehension.